Sketch2Style:
A Controllable AI Fashion Design Using Diffusion Models & Sketch Conditioning

Group 3:
Rupali Ravindra Shetye
Kartavya Mandora

Date: 2nd Apr 2026



LONG ISLAND UNIVERSITY

Source: https://www.freepik.com/premium-ai-image/aidriven-fashion-design-future-vision_83711218.htm

OUTLINES

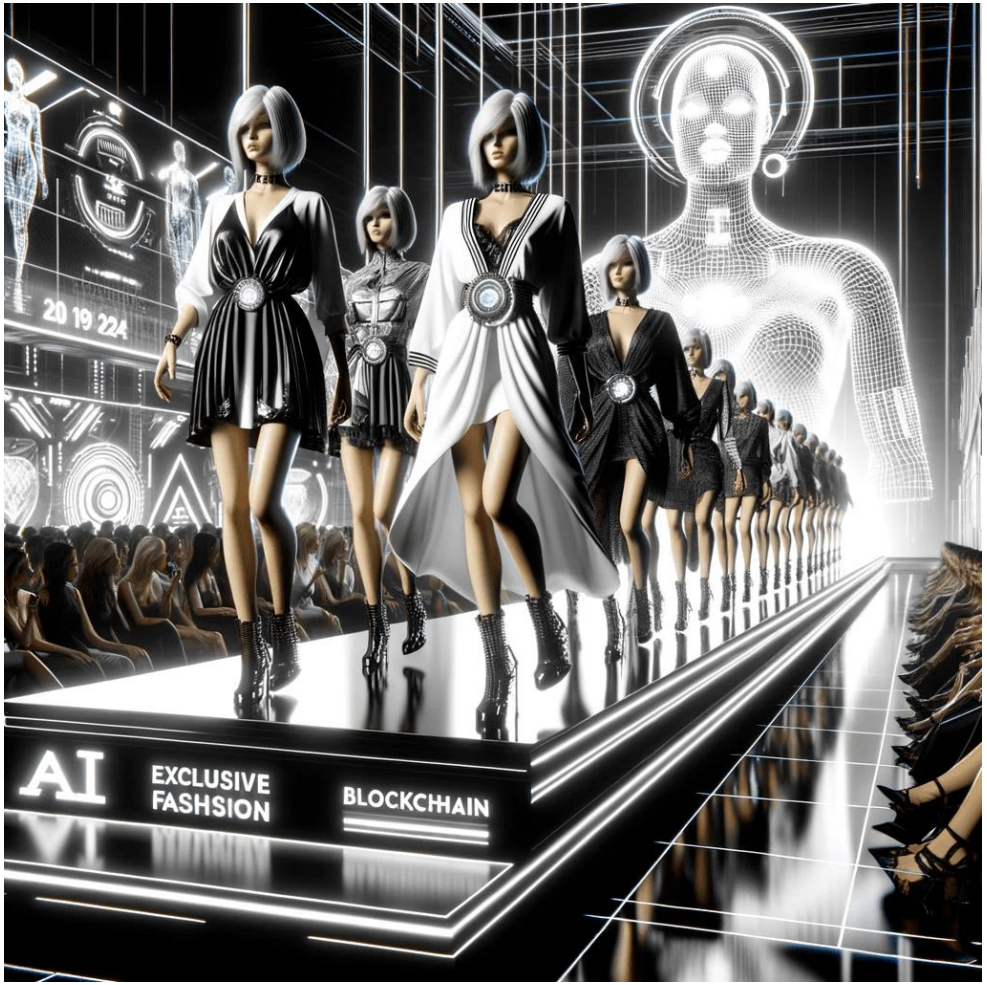
LIU

1. Introduction
2. Application
3. Literature Review
4. Literature Review Results
5. Methodology
6. Model Architecture
7. Dataset
8. Understanding the Functions
9. Training Strategy
10. Obtained Results
11. Conclusion & Future Work
12. Team Contribution
13. Timeline
14. References

INTRODUCTION

LIU

- Fashion design starts from sketches
- Designers need **structure + style control**
- Diffusion models generate realistic images
- BUT lack **spatial control**

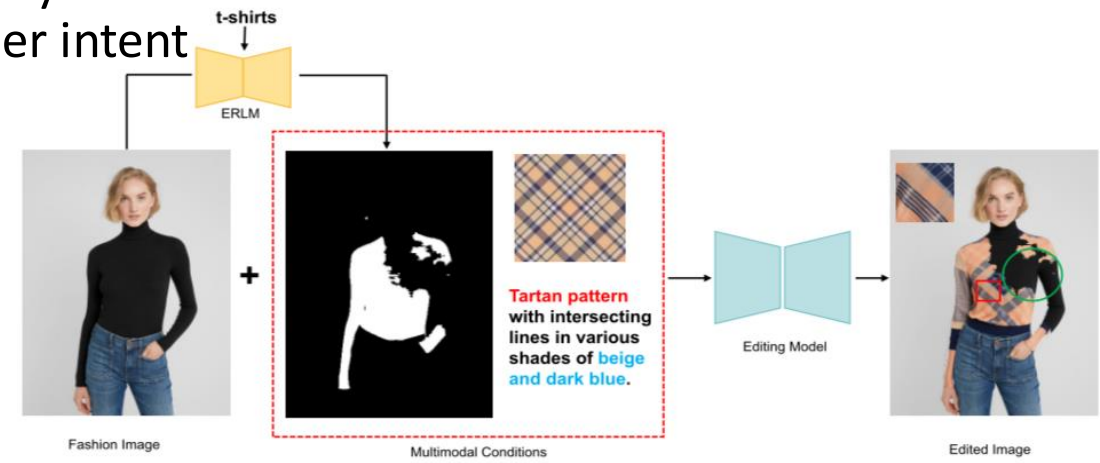


Source: https://davincinext.com/2024/01/22/ai-web3-fashion-revolution-202/

Problem Statement

Current AI tools suffer from:

- ✗ Cannot preserve sketch geometry
- ✗ Output not aligned with designer intent



Our solution directly addresses each gap:

- ✓ Sketch-conditioned generation using ControlNet
- ✓ Text prompts for style consistency
- ✓ Gradio interface for usability

Source: https://www.themoonlight.io/en/review/dpdedit-detail-preserved-diffusion-models-for-multimodal-fashion-image-editing

APPLICATION



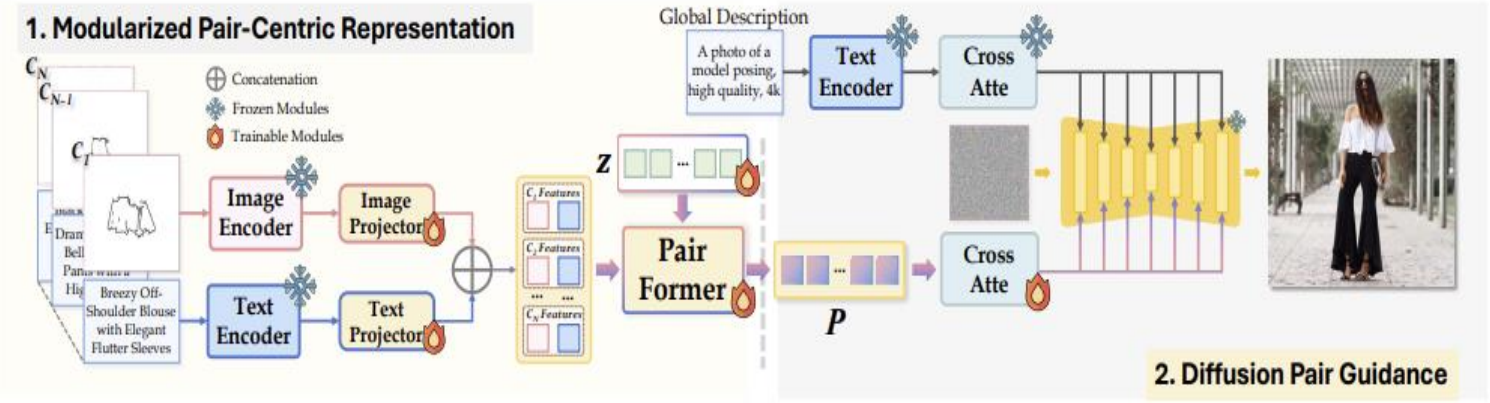
Source: https://www.mirrorsize.com/blogs/custom-fits-in-clicks-ais-role-in-made-to-measure-apparel

- AI-assisted fashion prototyping
- Rapid design exploration
- Cost reduction
- Support for small designers
- Creative exploration tool

LITERATURE REVIEW

Paper 1: LOTS (ICCV 2025)

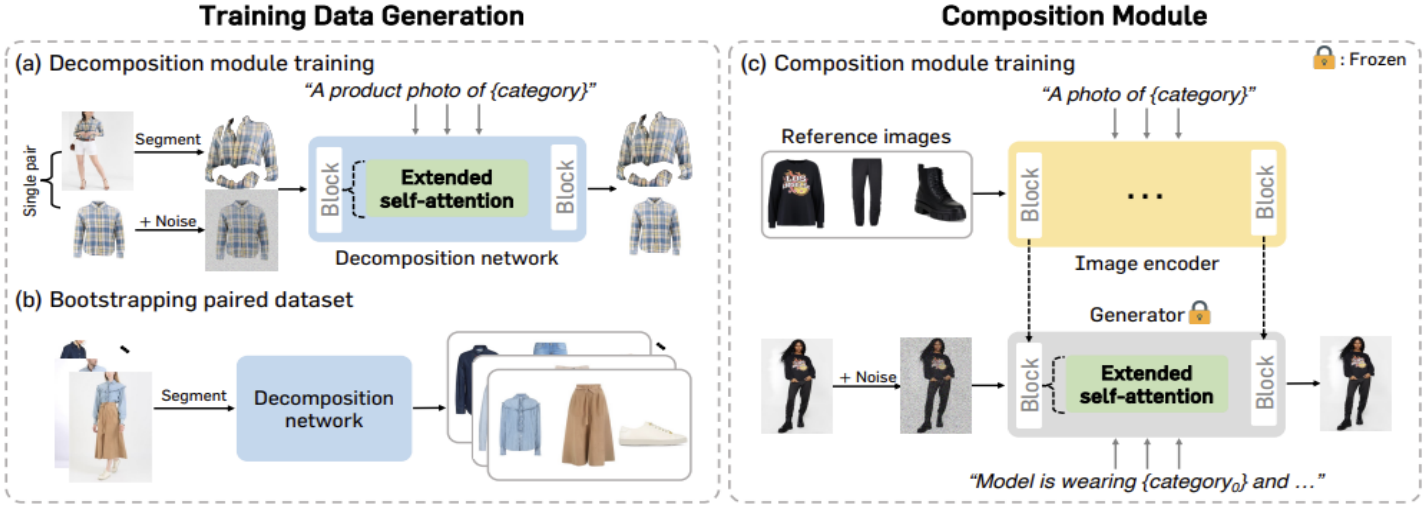
- **Title:** LOTS of Fashion
- Multi-conditioning diffusion
- Sketch + Text pairing
- Modular attention-based merging
- Introduced Sketchy dataset
- Improved attribute localization



Source: F. Girella et al.

Paper 2: BootComp (CVPR 2025)

- **Title:** Controllable Human Image Generation
- Multi-garment conditioning
- Two-stage pipeline
- Synthetic paired data generation
- Extended self-attention
- +30% improvement in MP-LPIPS(Multi-Scale Learned Perceptual Image Patch Similarity)

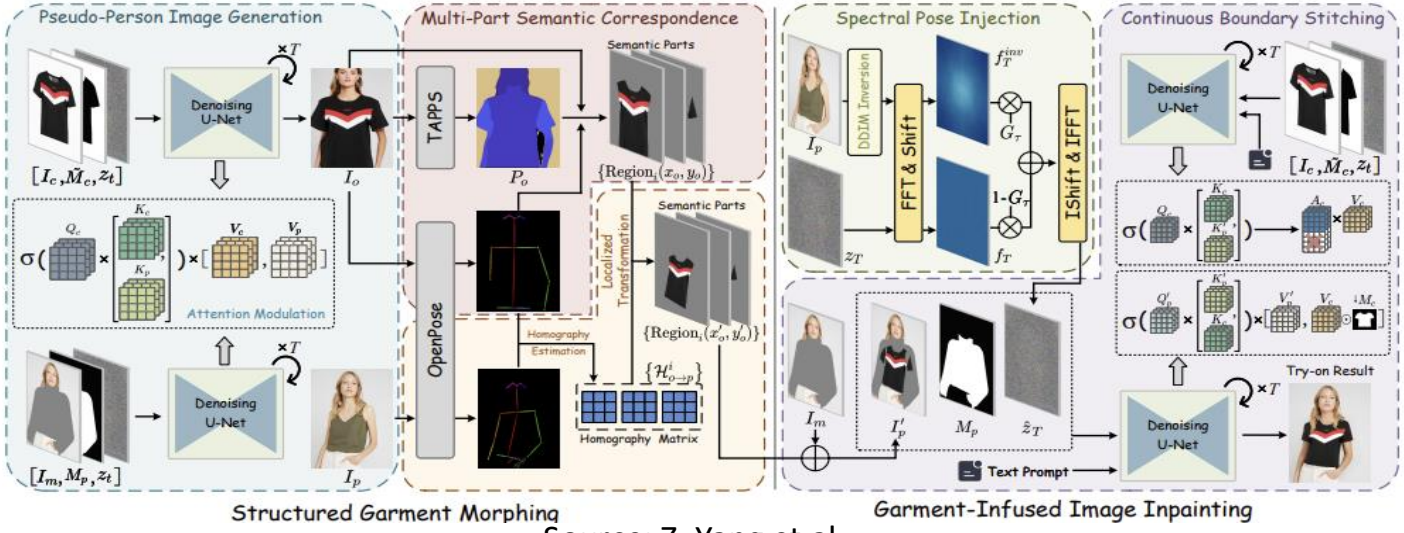


Source: Y. Choi et al.

LITERATURE REVIEW

Paper 3: OmniVTON (ICCV 2025)

- **Title:** Training-Free Universal Virtual Try-On
- Structured Garment Morphing
- Spectral Pose Injection
- Continuous Boundary Stitching
- Training-free approach
- Strong pose consistency



Source: Z. Yang et al.

LITERATURE REVIEW RESULTS

LIU

Aspect	LOTS	BootComp	OmniVTON
Objective	Sketch-text generation	Multi-garment control	Training-free try-on
Conditioning	Sketch + Text	Multi-garment images	Pose + garment
Training	Fully trained	Trained	Training-free
Strength	Layout accuracy	Texture fidelity	Pose consistency
Limitation	Complex design	Not sketch-driven	Not design-focused

Research Gap

- No ControlNet-based fashion sketch system
- No study using Fashion-MNIST
- No simple designer-friendly implementation
- Need spatial enforcement via ControlNet

Our contribution: First ControlNet-based fashion generation on Fashion-MNIST with accessible web interface

ALGORITHM/APPROACH/METHODOLOGY

LIU

Step 1 Dataset Exploration	Fashion-MNIST: 70,000 images, 10 clothing categories, Balanced dataset Split: 9,600 training, 2,400 validation, 2,000 testing
Step 2 Data Preprocessing	Resize 28×28 → 256×256, Convert grayscale → RGB, Normalize: [1,-1], Apply Canny edge detection, Generate sketch maps
Step 3 Text Prompt Construction	Convert labels to descriptive prompts Encode using CLIP Provide semantic control
Step 4 Model Pipeline	Edge map → ControlNet, Prompt → CLIP encoder, Random noise → Stable Diffusion v1.5, Dual conditioning during denoising, VAE decoder → Final image
Step 5 Training Strategy	Each sample includes: Target image, Edge map, Text prompt Training process: Add noise to latent, Predict noise residual, Keep base diffusion mostly frozen, Train ControlNet layers

OVERALL WORKFLOW

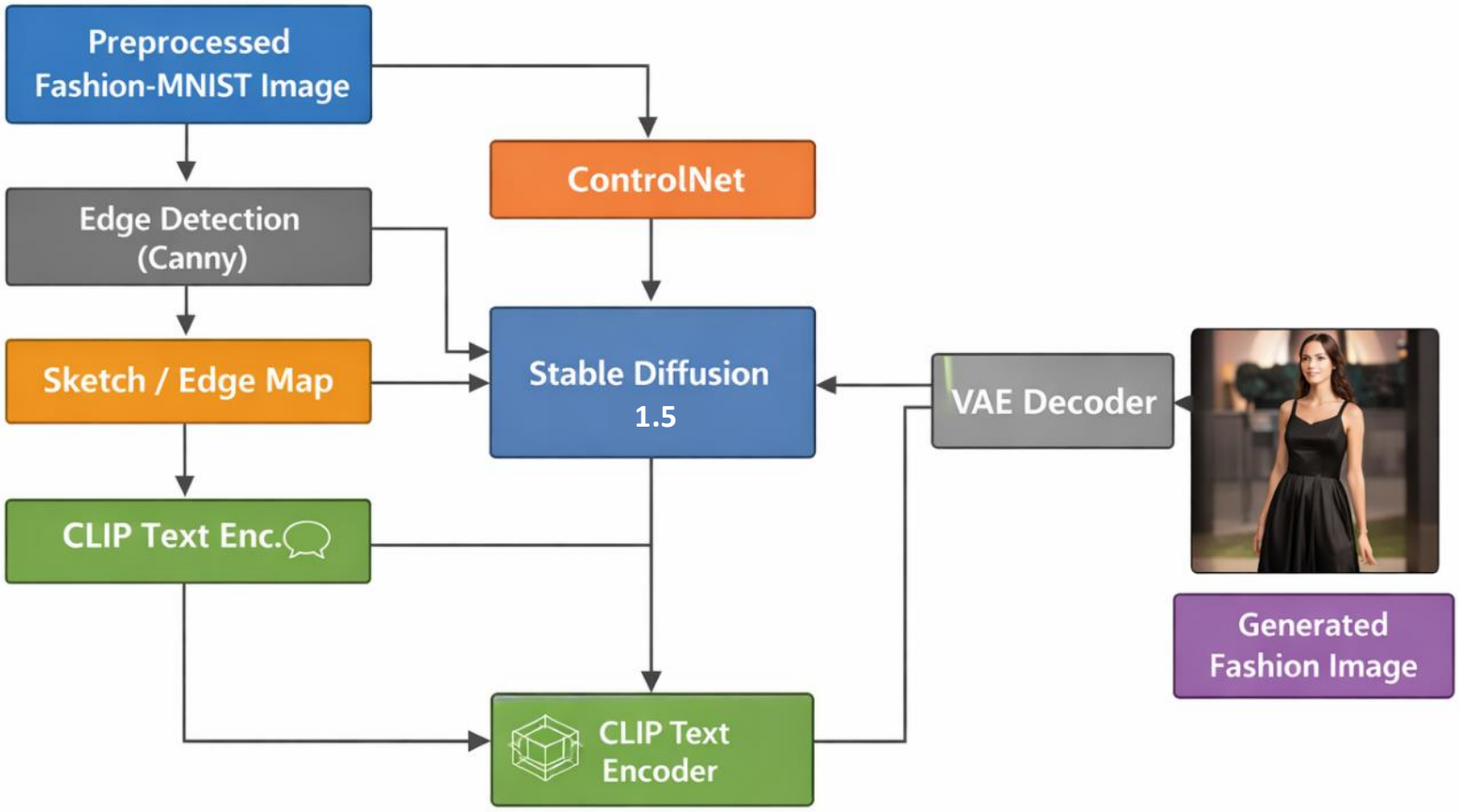


Figure 1: Overall Workflow

MODEL ARCHITECTURE

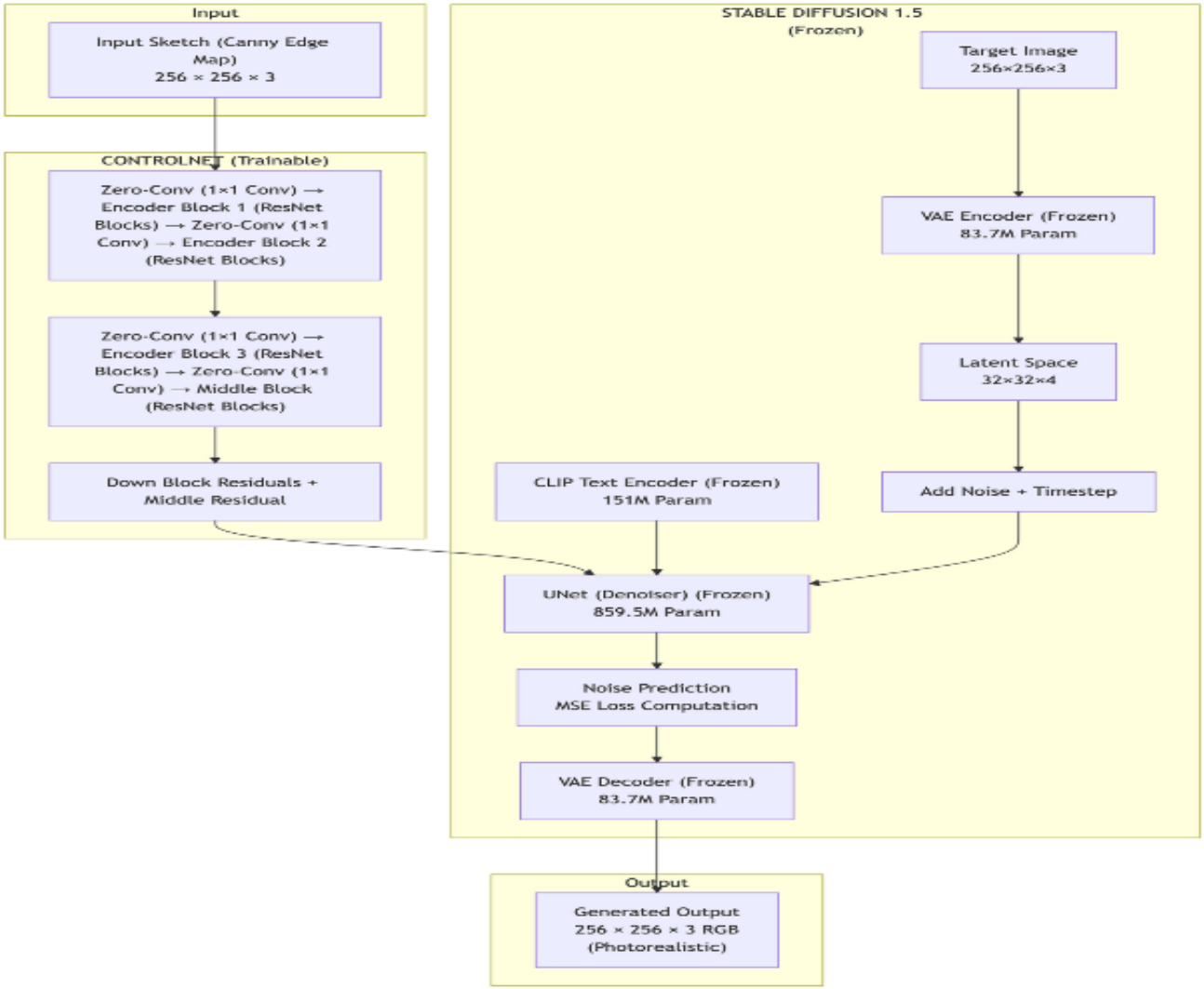


Figure 2: Model Architecture

DATASET STATISTICS

LIU

Dataset Name: Fashion-MNIST Overview

- Total Images: 70,000
- Image Type: 28 x 28 grayscale
- Categories: 10 balanced classes (7,000 each)
- Pixel Range: 0-255

Selected Categories for Midterm: Dress & Sneaker

DATASET DISTRIBUTION (SELECTED CATEGORIES)

LIU

Dataset Split After Filtering

Type	Samples
Training	9,600
Validation	2,400
Test	2,000
Total	**16,000**

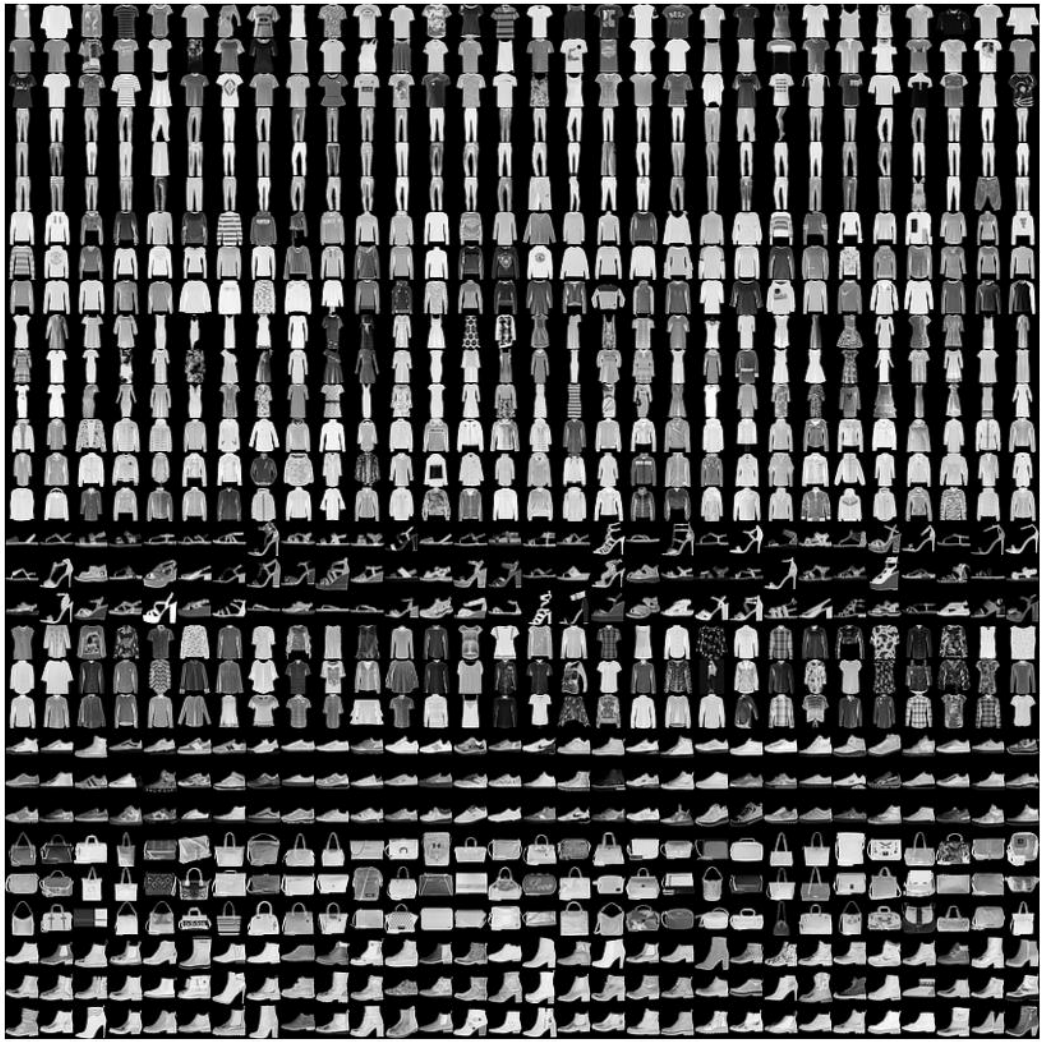
Why 5,600 instead of 7,000?

- Original Fashion-MNIST has 7,000 images per category
- 20% reserved for validation (1,400 per category)
- Remaining 80% used for training (5,600 per category)

Per Category Breakdown

Category	Training	Validation	Test	Total
Dress	5,600	1,400	1,000	8,000
Sneaker	5,600	1,400	1,000	8,000
Combined	**11,200**	**2,800**	**2,000**	**16,000**

DATASET EXPLORATION



Source: https://www.kaggle.com/code/gpreda/cnn-with-tensorflow-keras-for-fashion-mnist

PRE-TRAINED MODEL SELECTIONS

LIU

Stable Diffusion Versions

Version	Text Encoder	Resolution	Parameters	Key Characteristics
v1.4	CLIP ViT-L/14	512×512	860M	First widely released version
v1.5	CLIP ViT-L/14	512×512	860M	Improved image quality, best ControlNet compatibility
v2.0	OpenCLIP	768×768	865M	Larger text encoder, higher resolution
v2.1	OpenCLIP	768×768	865M	Refined version of v2.0

Selected: v1.5 because it offers best compatibility with ControlNet and lower computational requirements for T4 GPU training.

ControlNet Variants

Variant	Conditioning Input	Best Use Case	Selected
Canny	Edge maps	Sketch preservation, structural control	YES
Depth	Depth maps	3D structure preservation	NO
Pose	Human keypoints	Human pose control	NO
Scribble	Rough sketches	Freehand drawing	NO
MLSD	Line detection	Architectural drawings	NO
Normal	Normal maps	Surface detail control	NO

Selected: Canny variant because Fashion-MNIST edge maps closely match designer sketches and provide precise structural guidance for garment shapes.

MODEL PARAMETER SUMMARY

LIU

Trainable vs Frozen Parameters

Component	Parameters	Trainable	Status
Stable Diffusion UNet	859,520,964	0	Frozen
Stable Diffusion VAE	83,700,000	0	Frozen
CLIP Text Encoder	151,000,000	0	Frozen
ControlNet	361,279,120	361,279,120	Trainable
TOTAL	**1,455,500,084**	**361,279,120**	**24.8% Trainable**

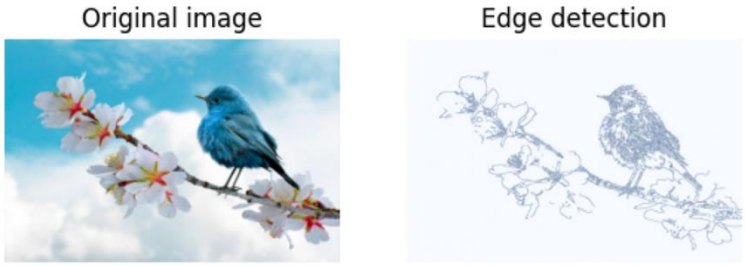
Key Insight: Only ControlNet (361.3M parameters) is trained. Base model remains frozen.

UNDERSTANDING THE FUNCTIONS

Role of Each Model

- **Stable Diffusion v1.5:**

  Generates realistic images with UNet



Source: https://medium.com/the-bytedoodle-blog/canny-edge-detection-deb0d62a8003

- **ControlNet:**

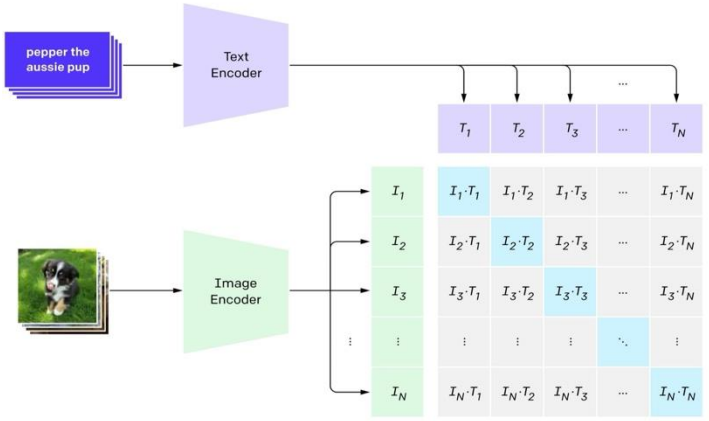
  Enforces sketch structure

- **CLIP(Contrastive Language-Image Pre-training):**

  Guides semantic meaning by text prompts

  Used Constant prompt:

  Ex: "a sneaker, fashion product photo, studio lighting, high quality"



- **VAE (Variational Autoencoder)**:

  Encodes and decodes images

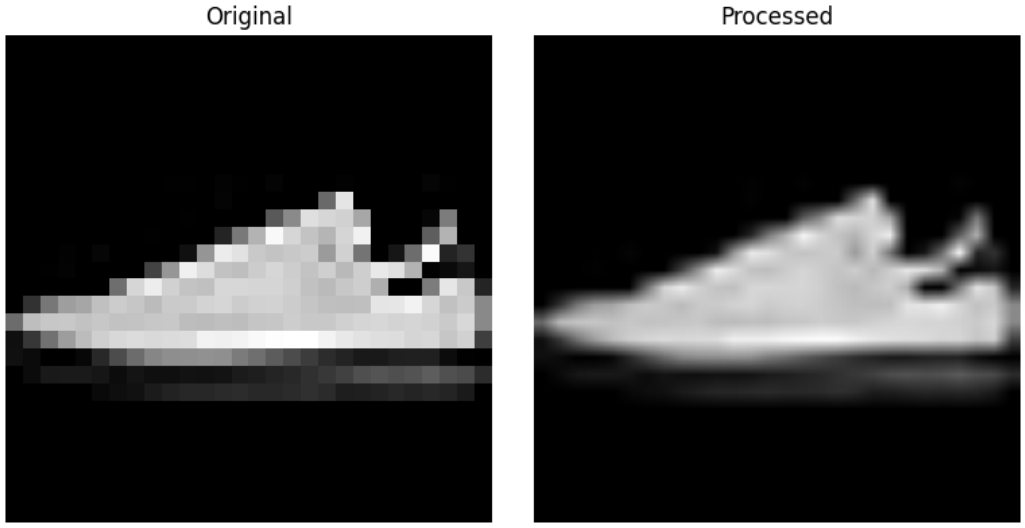
Source: https://viso.ai/deep-learning/clip-machine-learning/

TRAINING STRATEGY

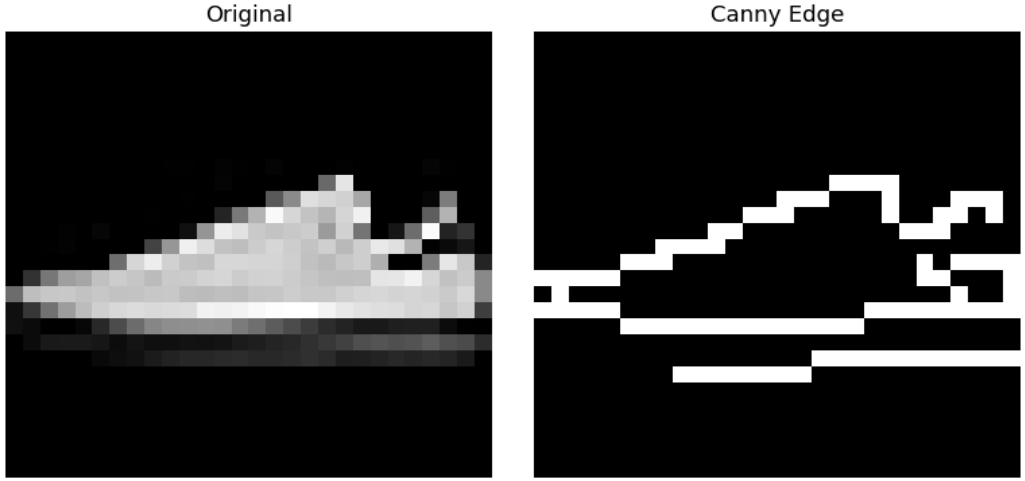
- Train only ControlNet
- Freeze:
  - UNet
  - VAE
  - CLIP
- Batch size = 2
- Steps ≈ 24,000 (for 5 epochs)
- Stable convergence

OBTAINED RESULTS

Preprocessing



Canny Edge Detection



OBTAINED RESULTS

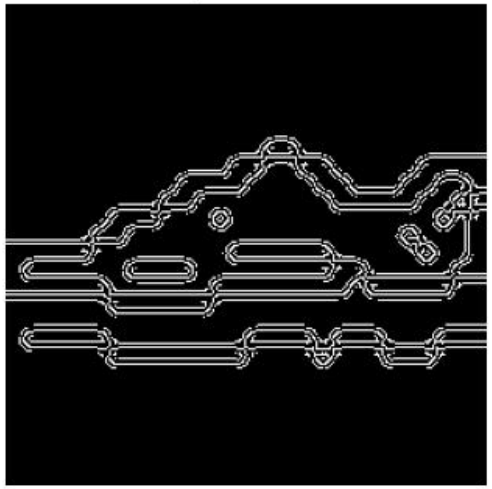
LIU

Training

```
Starting Epoch 1/5
Epoch 1/5 | Batch 200/4800 | Global Step 200 | Loss: 0.1419
Epoch 1/5 | Batch 400/4800 | Global Step 400 | Loss: 0.0849
Epoch 1/5 | Batch 600/4800 | Global Step 600 | Loss: 0.0446
Epoch 1/5 | Batch 800/4800 | Global Step 800 | Loss: 0.0277
Epoch 1/5 | Batch 1000/4800 | Global Step 1000 | Loss: 0.0178
Checkpoint saved at: checkpoints/controlnet_epoch_1_step_1000.pt
Epoch 1/5 | Batch 1200/4800 | Global Step 1200 | Loss: 0.0360
Epoch 1/5 | Batch 1400/4800 | Global Step 1400 | Loss: 0.0061
Epoch 1/5 | Batch 1600/4800 | Global Step 1600 | Loss: 0.0458
Epoch 1/5 | Batch 1800/4800 | Global Step 1800 | Loss: 0.0547
Epoch 1/5 | Batch 2000/4800 | Global Step 2000 | Loss: 0.0420
Checkpoint saved at: checkpoints/controlnet_epoch_1_step_2000.pt
```

Image Generation or Testing

Input Sketch



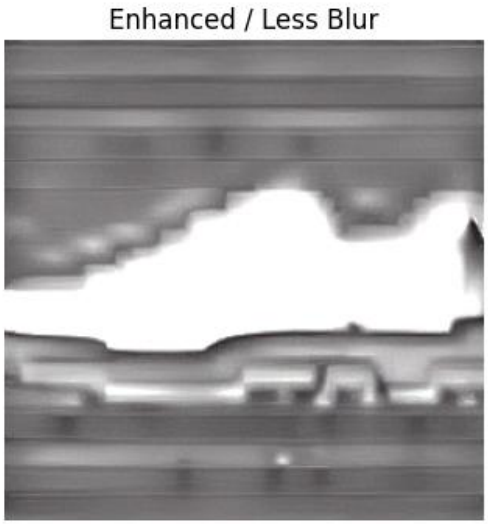
Generated Fashion Image



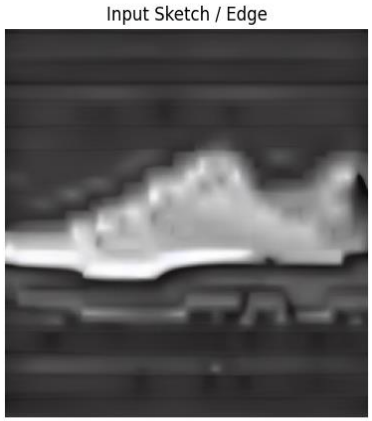
OBTAINED RESULTS

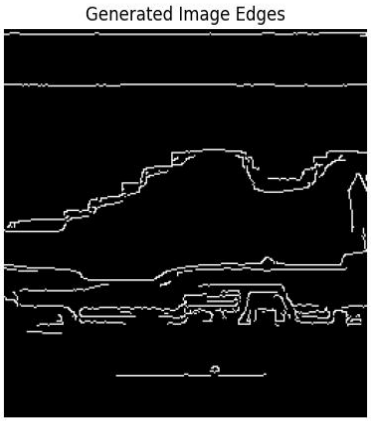
Sharpen Image

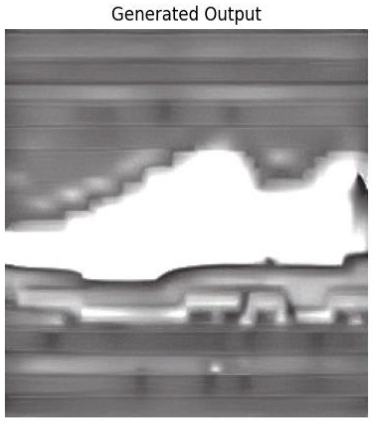




Evaluation



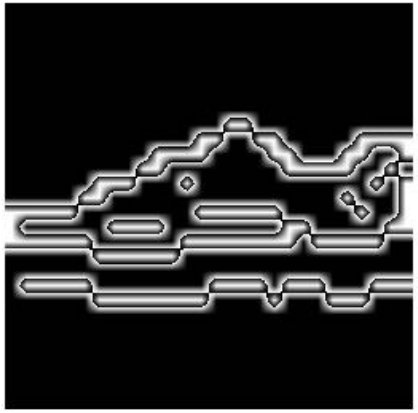




OBTAINED RESULTS

Comparison b/w Clean Sketch & Messy Sketch

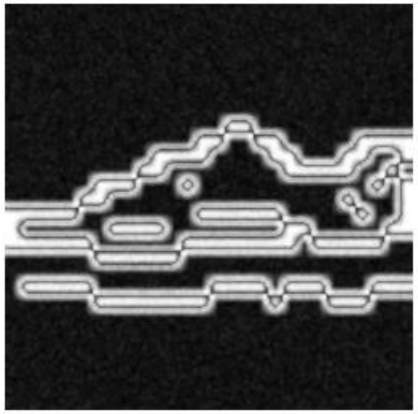
Clean Sketch



Output from Clean Sketch



Messy Sketch



Output from Messy Sketch



OBTAINED RESULTS

LIU

- **Training Result:**
  - Initial loss ≈ 0.12
  - Final loss ≈ 0.0025
  - No NaN issues
  - Stable convergence

- **Observations**
  - Preserves garment structure
  - Generates realistic shapes
  - Text prompt influence visible
  - Model remains stable
  - Recognizable output
  - Slight Quality Drop

Metric	Value
SSIM(Structural Similarity Index Measure)	-0.0024
CLIP Score(Contrastive Language-Image Pre-training)	0.1854
Inference Time	~2 sec

CONCLUSION & FUTURE WORK

LIU

➢ Conclusion:

- Successfully built sketch-conditioned system
- ControlNet enables structural control
- Pipeline works end-to-end

➢ Future Work:

- Try to implement more categories
- Increase resolution (512×512)
- Add FID metric
- Improve training time
- Add user sketch interface and human evaluation



Source: https://www.lezhougarment.com/ultimate-guide-to-3d-fashion-design-the-future-of-fashion/

ETHICAL CONSIDERATIONS

LIU

Aspect	Summary
Data Ethics	Public Fashion-MNIST dataset; no human subjects; 70,000 images across 10 categories
Model Ethics	Open source (MIT License); reproducible with fixed random seeds; balanced representation
Scientific Integrity	Results reported honestly; all prior work properly cited
Dual Use	Academic research only; assists designers, not replaces creativity
Environmental Impact	2.5 kg CO2 (equivalent to driving 10 miles)

TEAM-WORK DISTRIBUTION

Team Member	Primary Contributions	Responsibilities
RUPALI	• Project Lead & Architecture Design • Literature Review Coordination • Model Architecture Design • Final Report Compilation	• Led weekly team meetings and coordinated project timeline • Design ControlNet + Stable Diffusion architecture • Coordinate literature review across 10 papers • Integrate all sections into final proposal
KARTAVYA	• Data Acquisition & Preprocessing • Edge Detection Pipeline • Implementation Lead • Results Documentation	• Download and process Fashion-MNIST dataset • Implement Canny edge detection pipeline • Develop Gradio web interface • Document technical implementation details

TIMELINE

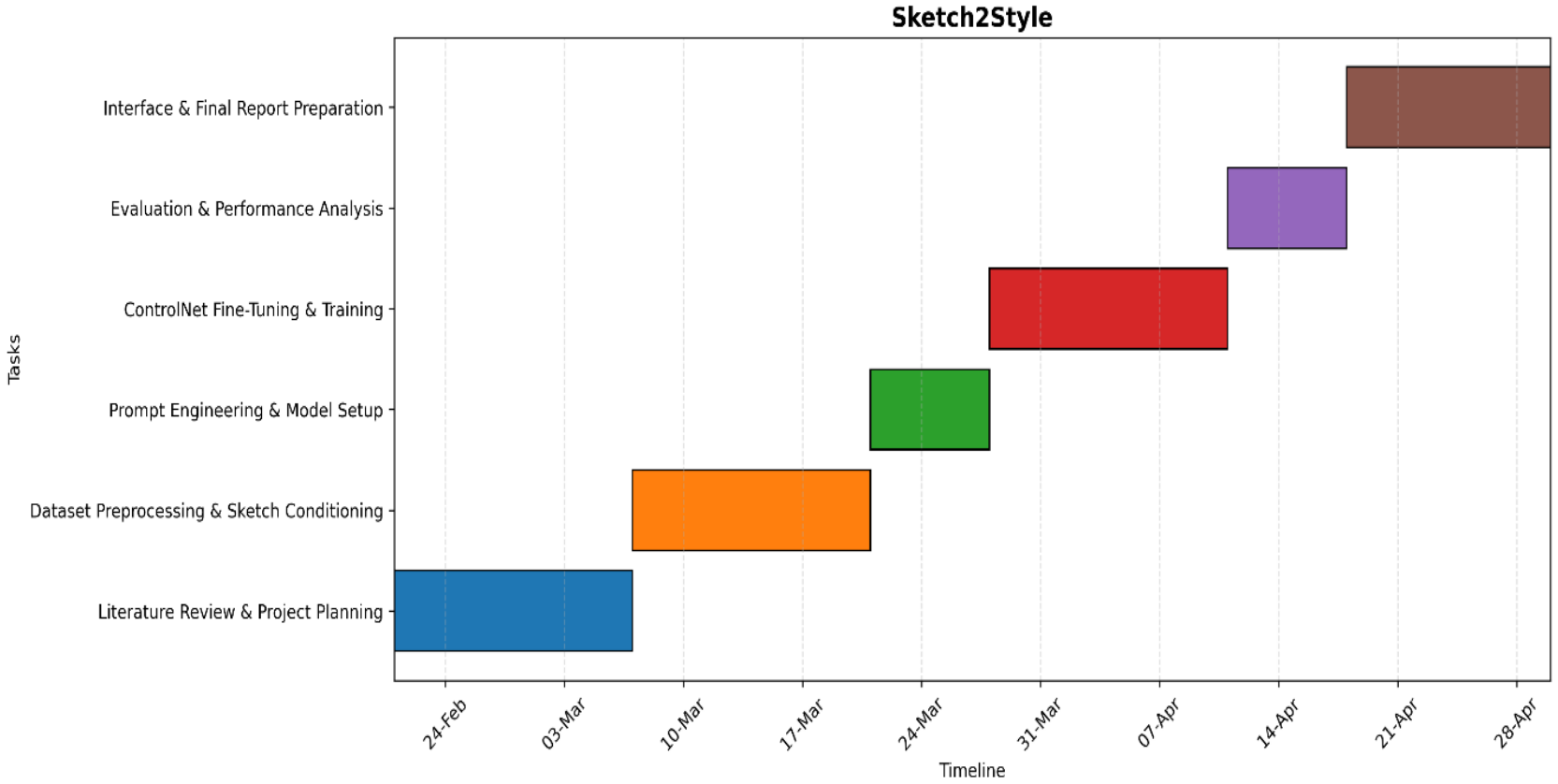


Figure 3: Gantt Chart

REFERENCES

[1] R. Rombach et al., "High-Resolution Image Synthesis with Latent Diffusion Models," CVPR, 2022.

[2] L. Zhang et al., "Adding Conditional Control to Text-to-Image Diffusion Models," ICCV, 2023.

[3] H. Xiao et al., "Fashion-MNIST: A Novel Image Dataset," arXiv, 2017.

[4] D. Podell et al., "SDXL: Improving Latent Diffusion Models," ICLR, 2024

[5] C. Mou et al., "T2I-Adapter: Learning Adapters for Text-to-Image Diffusion," AAAI, 2024.

[6] F. Girella et al., "LOTS of Fashion! Multi-Conditioning for Image Generation," ICCV, 2025.

[7] Y. Choi et al., "Controllable Human Image Generation with Personalized Multi-Garments," CVPR, 2025.

[8] Z. Yang et al., "OmniVTON: Training-Free Universal Virtual Try-On," ICCV, 2025.

Questions